Building Your Research

PS50008A Week 13 | Lecture

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Learning Outcomes

By the end of this lecture you will be able to:

- Operationalise abstract concepts into measurable variables
- Write clear hypotheses (one-tailed vs two-tailed) with justification
- Choose between-subjects or within-subjects designs and explain trade-offs
- Connect your design choice to the appropriate statistical test

Setting the Scene

Where We Are

Week 12: You learned the framework

- Hypotheses (H_0 and H_1)
- Independent and dependent variables
- Between-subjects and within-subjects designs

Week 13: Now we USE it

- Build YOUR Research Plan
- Make real decisions about real studies

The Research Plan Assessment

> **Due**: Friday 27th February (Week 16)
>
> **Weight**: 20% of module mark
>
> **Task**: Design an experiment on ONE of three topics

You must demonstrate you can plan rigorous research — even before collecting data.

The Three Topics

Choose ONE for your Research Plan:

1. **Font difficulty → Memory**
   - Does harder-to-read text improve memory?
2. **Hydration → Cognitive performance**
   - Does drinking water improve thinking?
3. **Eye contact → Perceived trustworthiness**
   - Does eye contact affect how trustworthy someone seems?

Act 1: From Concept to Measurement

Operationalisation (~25 min)

What is Operationalisation?

> **Operationalisation**: The process of turning an abstract concept into a concrete, measurable variable.

Your Research Plan requires you to specify:

- **Exactly** how you'll manipulate the IV
- **Exactly** how you'll measure the DV

Why Does It Matter?

- Vague variables = unmeasurable research
- Other researchers need to replicate your study
- Poor operationalisation = confounded results
- Reviewers (and markers!) will critique weak operationalisation

> **Bad**: "We measured happiness"
>
> **Good**: "Participants rated their current happiness on a 1-10 scale, with 1 = extremely unhappy and 10 = extremely happy"

Topic 1: Font Difficulty → Memory

> **The claim**: Reading in hard-to-read fonts creates "desirable difficulty" that enhances learning
>
> **Source**: Diemand-Yauman et al. (2011)

Questions to answer:

- What counts as "difficult to read"?
- How different should the fonts be?
- What "memory" outcome do we measure?

Operationalising “Font Difficulty”

Option 1: Fluent (Arial 12pt) vs Disfluent (Comic Sans 12pt, 75% grey)

Option 2: Fluent (Times New Roman 14pt) vs Disfluent (Haettenschweiler 12pt)

Option 3: Fluent (standard) vs Disfluent (Italicised, smaller font)

> **Note**
>
> The key is CLEAR CONTRAST between conditions that isolates font readability while keeping content identical.

Operationalising “Memory”

- **Free recall**: How many facts can participants remember?
- **Recognition test**: Multiple choice on material content
- **Cued recall**: Provide category, ask for specific items

Each measures slightly different aspects of memory. **Your choice must be justified**.

Topic 2: Hydration → Cognitive Performance

> **The claim**: Being hydrated improves cognitive function
>
> **Source**: Edmonds & Burford (2009)

Questions to answer:

- What does “hydration” mean operationally?
- What cognitive task(s) measure “performance”?

Operationalising "Hydration"

Option 1: Water group drinks 500ml; control group drinks nothing

Option 2: Water group drinks 250ml; control group sips (no swallowing)

Option 3: Measure time since last drink (observational)

> ⚠ **Potential Confound**
>
> If water group *believes* water helps cognition, expectation effects may drive results. Consider a placebo control?

Operationalising “Cognitive Performance”

- **Attention task**: Reaction time, sustained attention
- **Working memory**: Digit span, n-back task
- **Processing speed**: Symbol substitution
- **Combined score**: Multiple tasks averaged

Topic 3: Eye Contact → Trustworthiness

The claim: Eye contact increases perceptions of trustworthiness

Source: Argyle & Dean (1965); subsequent research on gaze and trust

Questions to answer:

- How do you manipulate eye contact?
- How do you measure “perceived trustworthiness”?

Operationalising “Eye Contact”

Option 1: Video of person with direct gaze vs averted gaze

Option 2: Photo with eyes meeting camera vs eyes looking away

Option 3: Duration manipulation: 2 seconds vs 6 seconds of eye contact

> **Note**
>
> Videos/photos provide control. Live interaction adds realism but introduces variability.

Operationalising "Trustworthiness"

- **Rating scale**: "How trustworthy does this person seem?" (1-7)
- **Behavioural proxy**: "Would you lend this person money?" (Yes/No)
- **Investment game**: How much would you invest with this person?
- **Multiple items**: Composite trustworthiness scale (Likert items)

Common Operationalisation Pitfalls

1. **Vague language**: "We changed the IV" — HOW exactly?
2. **Confounded manipulation**: Changing multiple things at once
3. **Unreliable measures**: Single-item scales, ceiling/floor effects
4. **Missing details**: Another researcher couldn't replicate

> **Test yourself**: Could a stranger run your study exactly from your description?

Act 2: Hypotheses That Work

One-Tailed vs Two-Tailed (~20 min)

Review: H_0 and H_1

> **Null Hypothesis (H_0)**: There is no effect or no difference
>
> **Alternative Hypothesis (H_1)**: There IS an effect or difference

We test H_0. If the data is very unlikely under H_0, we reject it and accept H_1.

Two Types of H_1

Two-Tailed (Non-directional)

$H_1: \mu_1 \neq \mu_2$

"There IS a difference"

Direction not specified

One-Tailed (Directional)

$H_1: \mu_1 > \mu_2$ OR $H_1: \mu_1 < \mu_2$

"Group 1 is higher/lower than Group 2"

Direction IS specified

When to Use Each

Two-Tailed: When you're uncertain about direction, or the effect could reasonably go either way

One-Tailed: When strong theory or prior evidence supports a specific direction

One-Tailed: Pros and Cons

Advantages:

- More statistical power to detect effect in predicted direction
- More precise research question

Disadvantages:

- Must JUSTIFY the direction (prior literature)
- Cannot claim significance if effect goes the OTHER way
- Some journals/reviewers are skeptical

Two-Tailed: Pros and Cons

Advantages:

- Safe choice when direction is uncertain
- Can detect effects in either direction
- More widely accepted

Disadvantages:

- Less statistical power
- Less precise prediction

Font Difficulty: Writing Hypotheses

Two-tailed:

- H_0: There is no difference in recall between fluent and disfluent font conditions
- H_1: Recall scores differ between fluent and disfluent font conditions

One-tailed:

- H_0: Disfluent font does not improve recall
- H_1: Participants in the disfluent font condition will recall more words than those in the fluent font condition

Justifying Your Direction

For your Research Plan, if you choose one-tailed:

1. Cite prior research supporting the direction
2. Explain the theoretical mechanism
3. Acknowledge what you'll conclude if the effect is opposite

> ⚠ **Assessment Requirement**
>
> Your Research Plan MUST include justification for your hypothesis choice.

Practice: Hydration Study

Write H_0 and H_1 for the hydration → cognitive performance study.

Would you choose one-tailed or two-tailed? Why?

Discuss with your neighbour (2 minutes)

BREAK

09:59

Act 3: Design Decisions

Between vs Within (~25 min)

The Fundamental Choice

> **Between-Subjects**: Different people in each condition
>
> **Within-Subjects**: Same people in all conditions

Both test the SAME hypothesis. The difference is HOW you assign participants.

Between-Subjects Design

How it works:

- Participant 1-30: Condition A
- Participant 31-60: Condition B
- Compare group means

Advantages:

- No order effects
- No practice/fatigue
- Simpler procedure

Disadvantages:

- Individual differences add noise
- Need more participants

Within-Subjects Design

How it works:

- All 30 participants do BOTH conditions
- Each person is their own control
- Compare within-person differences

Advantages:

- Controls individual differences
- More statistical power
- Fewer participants needed

Disadvantages:

- Order effects
- Practice/fatigue
- Demand characteristics

Decision Tree for Our Topics

Font difficulty → Memory

- Between: Group A reads fluent, Group B reads disfluent
- Within: Everyone reads BOTH (counterbalanced)

Which is better? What are the trade-offs?

Font Study: Design Analysis

Within-subjects concerns:

- If same content twice → memory for first reading confounds second
- If different content → is content equivalent?
- Order effects: Which font first?

Between-subjects concerns:

- Some people have better memory naturally
- Need twice as many participants

Hydration Study: Design Analysis

> **Think**: What would between-subjects vs within-subjects look like?
>
> What are the trade-offs for each?

- **Within**: Same person tested hydrated and dehydrated
  - Problem: Requires multiple sessions or dehydration manipulation
- **Between**: Different people in water vs no-water groups
  - Problem: Individual differences in cognitive ability

Eye Contact Study: Design Analysis

Vote: For the eye contact → trustworthiness study, which design would you choose?

a. Between-subjects (different photos/videos for each participant)

b. Within-subjects (same participant rates BOTH gaze conditions)

Within-subjects is common here: Each participant can rate multiple stimuli, and we compare their ratings across conditions.

Practical Constraints

Your design choice isn't just about statistics. Consider:

- **Time**: How long can you test each participant?
- **Participants**: How many can you realistically recruit?
- **Ethics**: Any concerns with repeated testing or manipulation?
- **Materials**: Can you create equivalent stimuli?

Act 4: From Design to Analysis

The Critical Link (~15 min)

Why Design Determines Test

> **The data structure from your design determines which statistical test is appropriate.**

Different designs produce different types of data relationships. The test must match.

Between-Subjects → Independent t-test

Design: 30 people in Font A, 30 different people in Font B

Data: Two independent groups of scores

Test: Independent samples t-test

Assumptions: Scores in Group A are independent of scores in Group B

Within-Subjects → Paired t-test

Design: 30 people rate trustworthiness for BOTH gaze conditions

Data: Paired scores (same person, two measurements)

Test: Paired samples t-test

Assumptions: Pairs of scores are related (same individual)

The Match Matters

⚠ Critical Point

Using the WRONG test gives you WRONG results.

- Independent t-test on paired data → loses power, wrong df
- Paired t-test on independent data → violates assumptions

For Your Research Plan

You MUST:

1. State your design (between or within)
2. State your test (independent or paired t-test)
3. EXPLAIN why this test matches your design

This is the “justify your choice of test” requirement.

Example Justification

> "This study uses a between-subjects design with different participants randomly assigned to each font condition. Because the data consist of scores from two independent groups, an **independent samples t-test** is the appropriate statistical test. This test compares the mean recall scores between the fluent and disfluent font groups to determine whether any observed difference is statistically significant."

Act 5: Method Section Walkthrough

What Goes Where (~15 min)

The Four Subsections

A Method section typically includes:

1. **Design**: What type? What are the IV and DV?
2. **Participants**: Who? How many? How recruited?
3. **Materials**: What stimuli, measures, equipment?
4. **Procedure**: What happened, step by step?

Design Subsection

> **Example**:
>
> "This study employed a between-subjects experimental design. The independent variable was font type (fluent vs disfluent), and the dependent variable was the number of words correctly recalled from a 20-word list."

Include: Design type, IV with levels, DV with how measured

Participants Subsection

> **Example**:
>
> "Sixty undergraduate psychology students (43 female, 17 male; mean age = 20.3 years, SD = 2.1) participated for course credit. Participants were randomly assigned to either the fluent font condition (n = 30) or the disfluent font condition (n = 30)."

Include: N, demographics, how recruited, how assigned

Materials Subsection

> **Example**:
>
> “The word list consisted of 20 common concrete nouns (e.g., ‘table’, ‘flower’). In the fluent condition, words were presented in 16-point Arial font. In the disfluent condition, words were presented in 12-point Comic Sans MS at 75% greyscale. A blank recall sheet was provided for participants to write remembered words.”

Include: Stimuli descriptions, measures, equipment

Procedure Subsection

> **Example**:
>
> "Participants were tested individually. After providing informed consent, they were presented with the word list for 60 seconds and instructed to memorise as many words as possible. The screen then went blank for 30 seconds (retention interval), followed by a 2-minute recall period during which participants wrote down all remembered words. Participation took approximately 10 minutes."

Include: Step-by-step what happened, timing, instructions given

The Replicability Test

> **Ask yourself**: Could another researcher, using only my Method section, run the exact same study?
>
> If not, what's missing?

Closing

Summary (~10 min)

Draft Results Table Structure

Your Research Plan asks for a **summary results table** (without data):

Condition	n	M	SD
Fluent Font			
Disfluent Font			

⚠ **“Do NOT invent data”**

You’re showing you know WHAT to measure and HOW to organise it — not making up numbers.

This Week: Your Tasks

1. **Choose your topic** (font, hydration, or eye contact)
2. **Operationalise your variables** precisely
3. **Write your hypotheses** with justification
4. **Decide your design** and explain trade-offs
5. **Match design to test** and justify

Research Plan Timeline

Key Dates

- **Week 13** (this week): Learn the content
- **Reading Week**: Write your draft
- **Week 16**: Research Plan due Friday 27 Feb

Start early. Get feedback from your lab instructor.

Key Takeaways

- **Operationalisation** makes abstract ideas testable
- **Hypothesis direction** requires justification
- **Design choice** involves trade-offs
- **Design determines test** — this link is essential
- **Method sections** must enable replication

Questions?

Questions?

See you in the lab for hands-on Research Plan work!